

Defining Informatics Competence for an AI-Enabled Clinical Workforce

What should every physician be able to do in a digitally mediated health system?

DISCLOSURE

No conflicts of interest to disclose.

The clinical environment has changed

Contemporary practice is increasingly mediated by information systems, algorithms, and data.

- | | | |
|-----------|--------------------------------|--|
| 01 | DIGITAL INFORMATION | EHR, imaging, patient-generated and longitudinal data |
| 02 | DIGITAL WORKFLOWS | Orders, documentation, communication, handoffs |
| 03 | DECISION SUPPORT | Alerts, predictive models, risk scores |
| 04 | ARTIFICIAL INTELLIGENCE | Ambient documentation, generative AI, predictive tools |
| 05 | PERFORMANCE DATA | Quality metrics, dashboards, outcomes, benchmarking |

The tools of clinical practice have changed faster than our definitions of clinical competence.

Informatics is no longer a specialty-only skill

Expertise in building information systems is different from competence in practicing within them.

CLINICAL INFORMATICS EXPERTISE

Design, implement, evaluate, and govern complex health information systems.

Specialized training

INFORMATICS COMPETENCE

Use information, digital systems, and computational tools safely and effectively in clinical practice.

Baseline clinical capability

Every physician does not need to become an informatician. Every physician increasingly needs informatics competence.

AI is already changing the work of physicians

The shift is visible in documentation, evidence retrieval, and the EHR itself.

AMBIENT AI

- 01** Listen
- 02** Interpret
- 03** Draft documentation

AI-MEDIATED KNOWLEDGE

- 01** Ask
- 02** Synthesize evidence
- 03** Verify and apply

AI INSIDE THE EHR

- 01** Summarize
- 02** Recommend
- 03** Predict and queue actions

The EHR is evolving from a passive repository into an active participant in clinical work.

Physicians increasingly interact with AI without explicitly deciding to “use AI.”

A COMPETENCY-BASED LENS

The question is not

“Does the physician know about AI?”

The question is

“What must the physician be able to do?”

Competence should be expressed as observable clinical capability, not familiarity with technology vocabulary.

Five domains of clinical informatics competence

Candidate capabilities for the contemporary physician, not requirements for becoming a clinical informatician.

- | | | |
|-----------|---|---|
| 01 | Data Literacy & Interpretation | Can I critically interpret the information informing care? |
| 02 | Digital Clinical Workflow | Do I understand how information and actions move through digital systems? |
| 03 | AI & Decision Support | Can I appropriately evaluate and act on algorithmic information? |
| 04 | Ethics & Responsible Informatics | Should this information or technology be used this way? |
| 05 | Learning From Clinical Data | Can I use data generated through care to improve future care? |

01 Data Literacy & Interpretation

Before clinicians can use AI well, they must understand the data that inform both human and algorithmic decisions.

UNDERSTAND

Where did the data come from?
What does it actually represent?

EVALUATE

Is it complete and accurate?
Is it fit for this purpose?

INTERPRET

What does the signal mean?
What uncertainty exists?

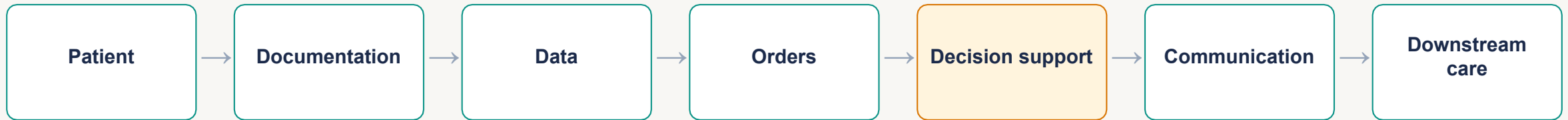
RECOGNIZE LIMITS

Missingness, measurement error, selection bias, derived vs observed data

Data are not synonymous with truth.

02 Digital Clinical Workflow

Clinical actions increasingly have computational consequences.



Competence means understanding where information originates, how it is transformed, where automation occurs, and how a digital action propagates through care.

Example: changing a diagnosis can affect coding, quality measurement, decision support, risk prediction, and future model training data.

03 AI & Decision-Support Competence

The competency is appropriate reliance, not model development.

01 What is this tool intended to do?

02 What information is it using?

03 How well does it perform, and in whom?

04 What might it be missing?

05 How much should its output influence this decision?

CONCEPTS CLINICIANS NEED

- Intended use
- Validation and generalizability
- Uncertainty and hallucination
- Automation bias
- Distribution shift
- Human oversight

Know when to trust, question, verify, or override algorithmic information.

Just because we can use information or AI... should we?

PRIVACY & CONSENT

Would the patient reasonably expect this use?

BIAS & EQUITY

Who might systematically benefit or be harmed?

TRANSPARENCY

When should AI involvement be disclosed?

AUTONOMY

Is technology supporting or constraining choice?

ACCOUNTABILITY

Who remains responsible for the decision?

COMMERCIAL INTERESTS

What is the system optimizing, and for whom?

Physicians need enough ethical competence to recognize when technology creates an ethical problem, and know how to respond.

05 Learning From Clinical Data

The contemporary clinician is simultaneously generating and consuming performance data.



- Interpret personal and team performance data
- Understand how quality metrics are constructed
- Recognize measurement limitations and unintended consequences
- Interpret longitudinal trends, not isolated measures
- Use data to identify opportunities for improvement
- Participate meaningfully in Learning Health System activities

LEARNING HEALTH SCIENCE

Routinely generated clinical data can be transformed into knowledge that improves subsequent care.

Data literacy includes the ability to learn from the data generated through clinical practice.

Surgery as an early test case

Our exploration of surgical informatics training exposed a broader question about baseline competence.

- | | | |
|-----------|--|--|
| 01 | A workforce gap | Surgeons practice in an increasingly data-intensive environment, but formal informatics training remains uncommon. |
| 02 | Multiple pathways for deeper training | Foundational courses, certificates, graduate degrees, Clinical Informatics fellowship, and team-science dyads. |
| 03 | The question that emerged | What informatics capabilities should every surgeon possess, regardless of whether they pursue formal informatics training? |
| 04 | A candidate baseline | Data literacy, digital workflow, AI and decision support, ethics, and learning from clinical data. |

These are candidate competencies for practicing surgery in a data- and AI-enabled healthcare system, not competencies for becoming a clinical informatician.

THE PROPOSITION



If these capabilities are required to practice medicine safely and effectively today,
they should be explicit within our definitions of physician competence.

Defining competence is only the first step.

How do we assess it?

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Clinical Informatics Is Clinical Competence

A cross-specialty, competency-based approach to workplace assessment, progressive entrustment, and intentional sampling

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Informatics is no longer adjacent to clinical care.

It is increasingly the medium through which care is performed.

That makes informatics a cross-cutting clinical competency—one that must be made visible, studied systematically, and assessed with intention.



Disclosure

I am co-founder of Firefly Lab and equity holder and board member of the Firefly Foundation.

These conflicts of interest are unrelated to this presentation.

Eight Milestone Frameworks Already Focus on Informatics

ACGME Milestones 2.0 frameworks, the competency is explicit in three and embedded in five.

3

of 8
frameworks

CLEARLY NAME INFORMATICS / DIGITAL HEALTH

Pathology — SBP4: Informatics

Diagnostic Radiology — SBP8:
Informatics

Internal Medicine — PC6: Digital Health

5

of 8 frameworks

EMBED INFORMATICS WITHIN EXISTING MILESTONE LANGUAGE

General Surgery — SBP / PBLI

Emergency Medicine — PC / SBP

Anesthesiology — PC / SBP

Family Medicine — SBP / PBLI

Obstetrics & Gynecology — SBP /
PBLI

**Same clinical work but different
label.**

THE SHARED COMPETENCY LENS

Interpret data · Navigate workflow
Create information that supports care · Learn from longitudinal performance

Four Discrete Skills Make Informatics Assessable



Interpret clinical data

Extract a clinically meaningful, source-aware signal from heterogeneous data. Recognize discordance and uncertainty.



Navigate digital workflow

Move safely across information systems and care transitions. Identify information loss, workflow limits, and closed-loop needs.



Create information that supports care

Document and communicate a reliable clinical representation that enables safe downstream decisions and handoffs.



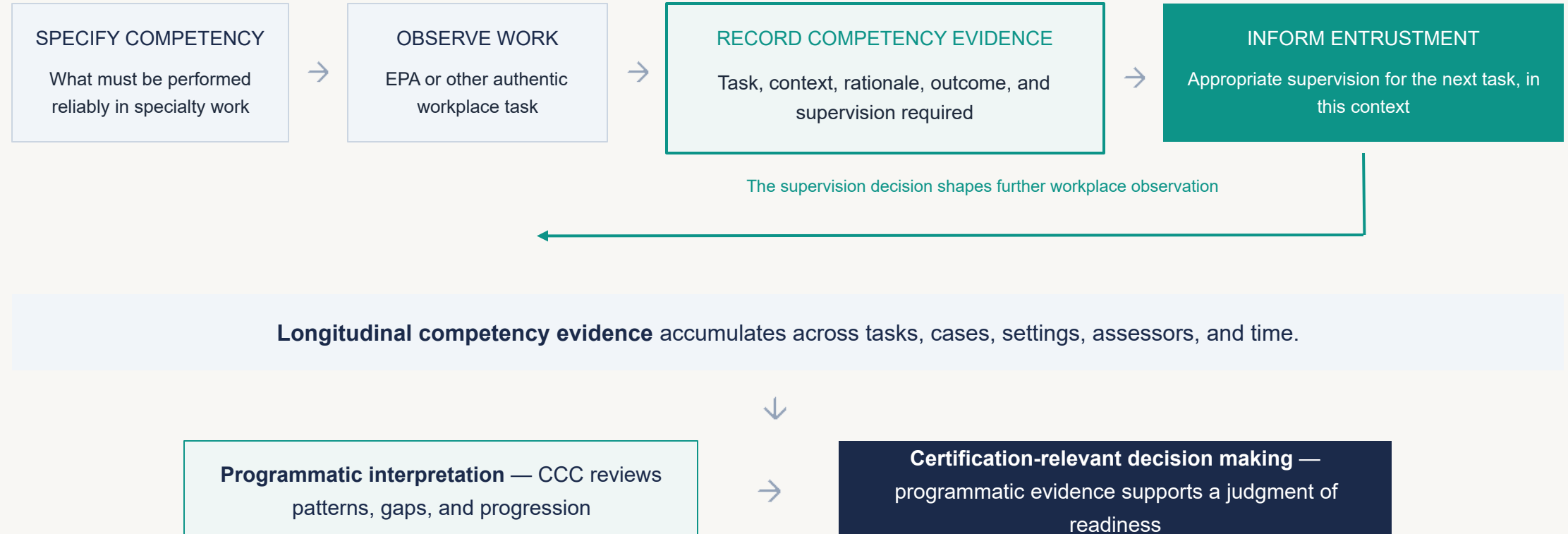
Learn from longitudinal performance

Use patient-, panel-, or system-level data to recognize variation, close care gaps, and improve clinical practice.

Competency-based design starts with naming the skill. Specialties then define the consequential clinical work in which that skill should be observed.

Workplace Evidence Informs Progressive Entrustment

Repeated clinical work makes competence visible & longitudinal evidence supports judgment. It is not a single-score proxy.



Competency is an enduring capability. Entrustment is a task- and context-specific supervision decision and both develop continuously through assessment and clinical work.

General Surgery: AI-Supported Reoperation Planning

A routine preoperative task becomes a high-value observation of embedded informatics competence.

ROUTINE RESIDENT WORK

Before a complex reoperation, the resident reconciles Epic, Care Everywhere, prior operative reports, and imaging. An illustrative AI-supported summary is reviewed—but verified against primary sources.

RETRIEVE

Cross-system record, imaging, and operative history

VERIFY

AI-supported synthesis against source records

PLAN

Source-aware strategy, uncertainty, and handoff

WORKPLACE EVIDENCE

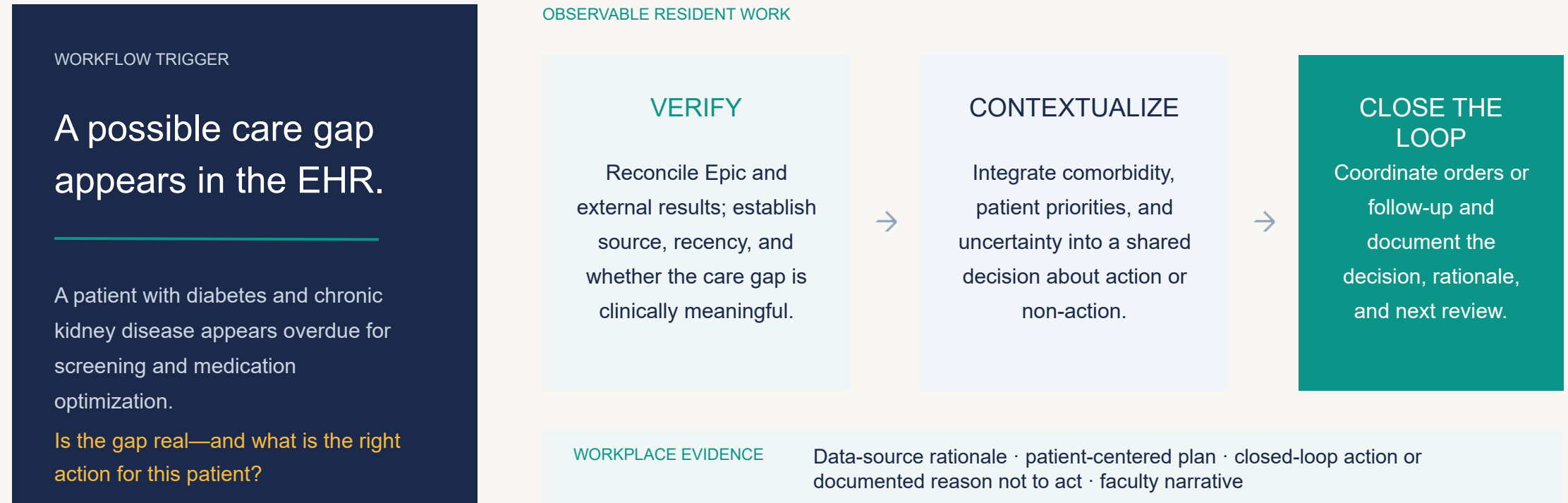
Source-aware problem representation · resolved or escalated discrepancy · verbal plan · faculty narrative on verification and supervision

Entrustment question: What supervision is appropriate for the next complex planning encounter with fragmented records and AI-supported input?



Internal Medicine: Closing a Care Gap With Longitudinal Data

A routine continuity-clinic task shows how EHR data becomes patient-centered clinical action.



Entrustment question: What supervision is appropriate for the resident's next EHR-supported, longitudinal care-gap intervention?

One Assessment “Grammar” Can Serve Different Specialties

The shared structure is not the tool or EPA label, but it is how a discrete clinical informatics skill becomes useful competency evidence.



THE SAME GRAMMAR, APPLIED TO SPECIALTY WORK

GENERAL SURGERY	Complex reoperation planning with fragmented records and AI-supported input	Verified problem representation + faculty narrative
INTERNAL MEDICINE	EHR-supported identification and closure of an actionable care gap	Source rationale + closed-loop plan + faculty narrative

An EPA is not the competency. It is a structured opportunity to observe a competency in consequential clinical work.

Intentional Sampling Determines Evidence Quality

Not every observation reduces uncertainty. The next assessment should be chosen for the information it is likely to add.

PASSIVE / CONVENIENCE SAMPLING

Observe what happens to be easy to assess.

Assessment concentrates in familiar tasks and among highly active faculty.

Complex, uncertain, or atypical work may remain unobserved.



Choose the observation that reduces uncertainty

UTILITY-AWARE SAMPLING

Observe where the evidence adds the most value.

Use workflow triggers to find authentic opportunities.

Use prior evidence to prioritize observations with the highest marginal information value.

UTILITY AS A DESIGN PRINCIPLE

$$\text{Utility} = \text{Reliability} \times \text{Validity} \times \text{Educational Impact} \times \text{Acceptability} \times \text{Cost}$$

Intentional assessment is not more assessment. It is a program of observation designed to produce credible evidence for progressive entrustment.

The Surgical Literature Shows That the Signal Is Designed

Evidence used for competence and entrustment reflects the assessment system and not only the learner's performance.

60.8%

FACULTY PARTICIPATION CONCENTRATION

of EPA submissions came from the most active 10% of faculty across 46 institutions.

52.8%

EPA-OPPORTUNITY CONCENTRATION

of 17,245 assessments involved only 4 of 18 EPA types across 37 institutions.

2.72 → 2.65

SAMPLING-SOURCE DIFFERENCE

Mean research rating, manual versus EHR-triggered assessments; **p = 0.005**.

Workplace assessment does not fail because it is workplace assessment.
It fails when taxonomy, sampling, and implementation are left to chance.

For clinical informatics, build the signal intentionally—from specialty work, workflow triggers, and information value.

A Cross-Specialty Goals Can Build This Deliberately

1

Map the work

Identify informatics competence already embedded in specialty practice.

2

Specify the skill

Translate hidden work into discrete, specialty-owned competency statements.

3

Prototype workplace evidence

Co-design tasks, triggers, and evidence standards with learners and faculty.

4

Learn from the signal

Test utility-aware sampling and whether evidence supports credible entrustment.

Informatics is everywhere in clinical work.
Its assessment should be present.

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Pathways to Informatics Competence

Fellowship depth, embedded breadth, and what pathology already shows about integrating informatics into training

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Disclosures

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My Pathway: Real, But Not Reproducible



01 NAVAL AVIATOR

P-3 Orion instructor pilot and mission commander.

02 PHYSICIAN

A career change into medicine.

03 SURGEON

General surgery residency at OHSU.

04 INFORMATICIAN

NLM T-15 informatics fellow; clinical informatics fellow; MS, biomedical informatics.

A real pathway. It should not take a career change and a dedicated fellowship to produce informatics fluency.

The competency is defined.
The methods exist.

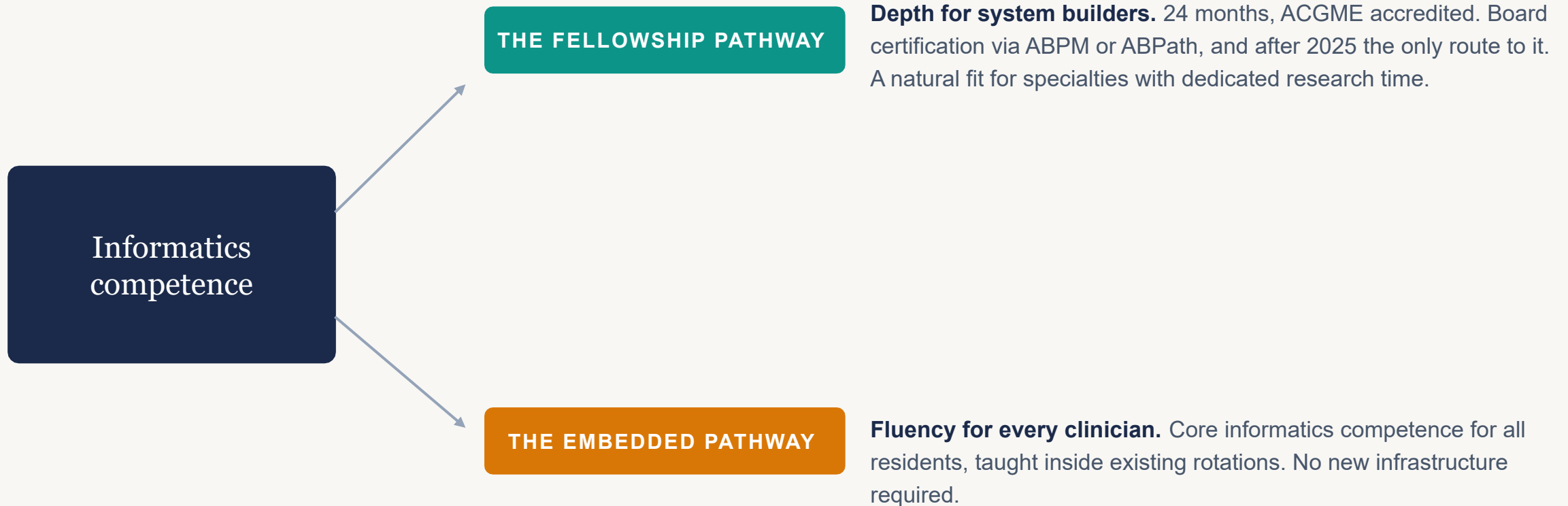
The remaining question is how competence reaches a
million physicians.

Two pathways already exist: fellowships that build depth, and embedded training
that builds fluency. Neither has to be invented from scratch.



Two Pathways Already Exist

Dr. Doberne separated expertise from competence. Each already has a route.



Different depth, the same destination: a workforce fluent in the systems through which care is delivered.

Informatics Is Already in the Resident's Day

The embedded pathway teaches what trainees are already doing.

Order entry and cross-system chart synthesis

Every plan starts with retrieving, weighing, and acting on digital data.

Mobile EPA microassessments

Workplace assessment arrives on the phone, inside the workflow.

Personal performance dashboards

Residents already read their own longitudinal data.



Naming these as competencies costs nothing. The work is already happening.

The Five Domains Already Have Classrooms

Dr. Doberne's domains map onto work residents do every rotation. The embedded pathway names and teaches them there.

01	Data Literacy & Interpretation	Chart review and pre-rounding: verifying the source, recency, and fitness of the data behind the plan.
02	Digital Clinical Workflow	Order entry, documentation, and handoffs: clinical actions with computational consequences.
03	AI & Decision-Support Competence	CDS alerts and AI-drafted notes: appropriate reliance taught at the moment of use.
04	Ethics & Responsible Informatics	Consent, privacy, and documentation practices on every service.
05	Learning From Clinical Data	M&M, case logs, and personal dashboards: residents reading their own longitudinal data.

These are clinical skills, not IT skills. The classrooms already exist; the curriculum just has to claim them.

The Groundwork Already Exists

A recognized subspecialty, named milestones, and embedded curricula are already in place.



NAMED

Several milestone frameworks already name informatics explicitly, as Part 2 showed; five more embed it.

TAUGHT

Embedded curricula exist as references; pathology's PIER maps national content to its milestone for all residents.

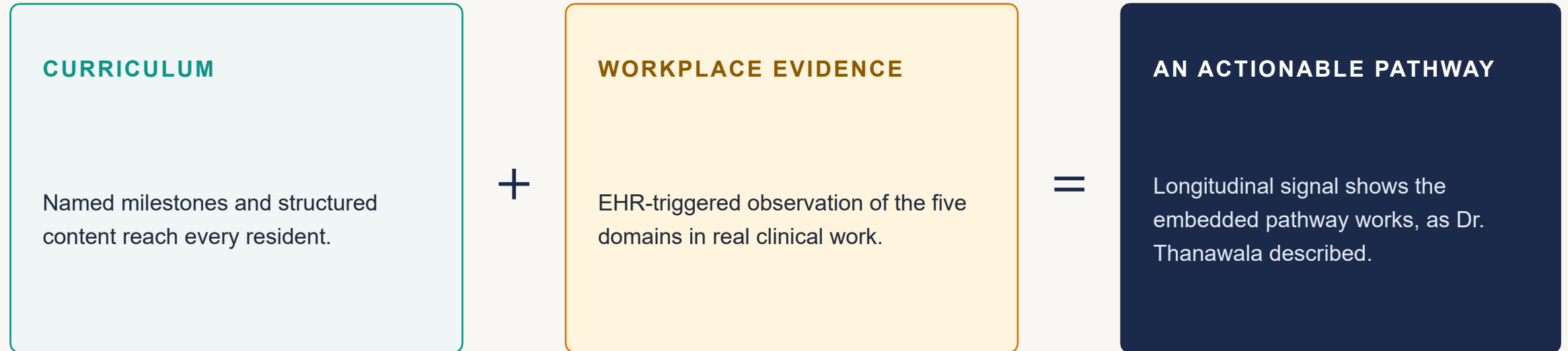
CREDENTIALLED

Board certification runs through ABPM for most specialties, and through ABPath for pathologists.

The gap is adoption, not precedent. The scaffolding exists. What remains is embedding the competencies in each specialty and making them visible.

Curriculum Names the Skill. Assessment Makes It Real.

Embedded curricula tell programs what to teach. Workplace evidence shows whether it was learned.



This is not a call for more assessment. It is the minimum evidence needed to show the embedded pathway works.

The Fellowship Pathway Is Now the Only Front Door

After the 2025 application cycle, ACGME fellowship is the sole route to board certification.

2014

First diplomates certified. Clinical informatics became an ABMS subspecialty, certified through ABPM and ABPath.

~2,000

Physicians certified to date. Most entered through the practice pathway, which closes after the 2025 application cycle.

24 mo

The remaining route. An ACGME-accredited fellowship. Specialties with dedicated research time already hold the space, surgery included.

Dedicated research years are the natural berth. Surgery's development years can absorb 24 months without lengthening training. The on-ramp exists; point trainees toward it.

Any Specialty Can Build Both Pathways



Both pathways already exist. The groundwork is laid.

What would it take in your specialty?

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